

TPI Sector Productivity Scorecards

Drivers and Indicators

Methodology and Data sources

By Olga Menukhin, William Sarsfield and Nathan McKeogh

The TPI's UK Sector Productivity Scorecard provides a comparative assessment of labour productivity performance by UK industries (e.g. manufacturing, services sectors, etc). It helps the TPI's audience assess the relative productivity drivers and drawbacks of growth (or lack of it), helps to identify challenges that need to be addressed and sources of growth in industries to engage with the business community. Visualising UK productivity data on the level of economic sectors can support different perspectives and analysis of labour productivity in the UK.

The scorecard evaluates four productivity driver categories each representing a cluster of economic indicators pertinent to productivity performance of sectors as shown in Figure 1:

- Productivity Measures
- Direct Labour Inputs: Employment, Skills & Wellbeing
- Direct Capital Inputs: Capital & Investment
- Indirect Drivers: Innovation, Technology, Market Exposure, other structural dynamics (could be added in the future)



Figure 1. Sector Productivity Drivers

The scorecard's structure is based on the production function perspective whereby a production function establishes a quantifiable relationship between physical inputs such as labour and capital, and physical output, i.e. output Y as a function of Direct Inputs: Labor L and Capital K , and total factor productivity (TFP) which captures the impact of other factors like technology, innovation, other organisational efficiency-related and TFP-enhancing forces. Being indirect drivers, the latter are key to explaining productivity differences. Tracking productivity and sector data using the production function logic enables clearer policy targeting, e.g. skills policy vs export policy vs technology or innovation diffusion. It also allows comparison of sector profiles, for instance high labour input and low innovation indicate a productivity lag.

Table 1. Direct/Indirect Scorecard logic

Type	Elements included	Economic role
Direct Inputs	Labour, Capital, Hours, Skills	Tangible, measurable inputs
Indirect Drivers	Innovation, Technology, Market exposure, and Structural dynamics	Enhance TFP, affect input quality or efficiency

The use of the Direct/Indirect logic to construct the TPI UK Sector Productivity scorecard makes it:

- Focused on productivity driver categories that reflect real-world economic levers
- Balanced between inputs, performance, and strategic context
- Modular for easier updates and expansion

This structure allows for analysing both productivity levels and drivers in a coherent way, with a view toward improving performance through policy, innovation, or workforce investment.

Productivity and Sector performance indicators included in each of the categories were collected from the UK Office for National Statistics (ONS). The inclusion of economic indicators was determined by consistent availability of statistics datasets by economic industry. Latest national statistical data are used for all indicators across the UK economy sectors. This means that year end varies per indicators. We use the [UK Standard Industrial Classification \(SIC\) Hierarchy](#) by the Office for National Statistics (ONS) to define UK economic sectors, see Table 2. The scorecards are generated for 19 sectors.

Table 2. Standard Industrial Classification (SIC 2007) sector codes and descriptions

	Sector	Section code	Division code
	Whole economy	A to T	01 to 98
1	Agriculture, forestry and fishing	A	01 to 03
2	Mining and quarrying	B	05 to 09
3	Manufacturing	C	10 to 33
4	Electricity, gas, steam and air conditioning supply	D	35
5	Water supply; sewerage, waste management and remediation activities	E	36 to 39
6	Construction	F	41 to 43
7	Wholesale and retail trade; repair of motor vehicles and motorcycles	G	45 to 47
8	Transportation and storage	H	49 to 53
9	Accommodation and food service activities	I	55 to 56
10	Information and communication	J	58 to 63
11	Financial and insurance activities	K	64 to 66
12	Real estate activities	L	68
13	Professional, scientific and technical activities	M	69 to 75
14	Administrative and support service activities	N	77 to 82
15	Public administration and defence; compulsory social security	O	84
16	Education	P	85
17	Human health and social work activities	Q	86 to 88
18	Arts, entertainment and recreation	R	90 to 93
19	Other service activities*	S	94 to 96
20	Activities of households as employers*	T	97 to 98

*A combined scorecard is generated for sectors S&T.

Measuring Growth

Short and long-term growth rates in the sectoral scorecards are calculated as simple percentage changes between the latest available year and the base year for each indicator. This approach captures the actual observed change in each variable over time, including temporary shocks such as the COVID-19 pandemic.

Using simple percentage change also ensures consistency across sectors, since the underlying dataset provides the same period coverage for each industry. The selected base year (typically 2019) follows ONS

guidance to use pre-pandemic benchmarks for productivity comparisons. Growth rates are interpreted as follows:

- change over time > 0.5 % - sectors show growth and are considered to have better performance
- change over time within ± 0.5 % - sectors are considered to have no significant change
- change over time < -0.5 % - sectors show a decrease in performance

This method allows for a clear and policy-relevant view of each sector's recent and longer-term productivity developments, highlighting where structural challenges - such as underinvestment, skills, or innovation - may be constraining growth.

If the long-term trend is not presented, this means that the ONS data is unavailable (a grey dot). For indicators expressed as a percentage such as the proportion of high-skilled roles, the sickness absence rate, and R&D intensity, change over the period is measured in percentage points rather than percentage change. This gives a more accurate reflection of the sector's growth or decline in that indicator. This avoids the distortion that can occur when percentage change is calculated on already-percentage figures, for example, a rise from 10% to 15% is a 5-percentage point increase, but a 50% relative increase.

Current Prices – Chained volume measures

Rows marked with a single asterisk (*) use their current prices in the sector value column but use the chained volume measures in the growth calculation. This is so that the values used to calculate growth are inflation adjusted and are correctly comparable with other growth metrics.

Indicators marked with N/A in the UK median and sector value columns are index values, these values on their own do not provide the user with any useful information and have therefore been removed.

Reversed (Inverted) Scale Indicators

For a small number of indicators, such as 'Low Skills', 'Annual Sickness Rate', 'Nominal Unit Labour Costs', 'Cost of Capital' and 'Job Intensity of Output', the relationship with productivity is inverse. In these cases, lower values indicate stronger productivity performance. The colour coding is therefore reversed to preserve interpretive consistency. For example, a sector with a 'Low Skills' growth value that is less than -0.5% would be coloured in green, since a decrease in low skills relates to higher levels of productivity. Indicators displayed on a reversed scale are marked with double asterisk (**). This convention allows for consistent comparison across the scorecard, ensuring that "improvement" is always represented by a movement in the same visual direction.

Indicator Ranking

The rightmost column displays each indicator's rank, showing where the sector's long-term growth rate places relative to all 19 UK sectors. Ranks are coloured using the same scheme as the long-term growth column, providing a consistent visual signal across both columns. Together, they allow the reader to assess not only whether a sector is growing or declining, but how that trajectory compares to the rest of the UK economy.

Datasets

This methodology document outlines the productivity output metrics and the economic indicators included within each driver category. All indicators used in the scorecard are sourced and reported consistently by

the ONS. Where data are unavailable at the SIC level, for reasons such as confidentiality or statistical disclosure, the indicator is shown in grey.

Productivity Output

This category comprises five indicators that assess and contrast the relative performance of labour productivity across different industries in the UK with respect to the entire UK economy, while a sector share in the UK GVA provides macroeconomic context by indicating the relative economic size and weight of a sector which can be useful for benchmarking and policy targeting.

Indicator	Description	Source/ Release
Labour Productivity (Output per Hour)	Output per Hour (OpH) worked by industry, £/h Gross value added (GVA) per hour values show how productivity in each industry compares to other industries, and the whole economy. Current price (CP) values are used for current year comparison (UK median and sector value columns). Chain volume measures (CVM) are used for comparison of growth (short-term change and long-term change columns). Time period: 1997-2025 Dataset: Output per hour worked, UK	ONS / 19 May 2026
Labour Productivity (Output per Job)	Output per Job (OpJ) by industry, £ ONS calculates output per job for each industry by dividing GVA by jobs for that industry. Current price (CP) values are used for current year comparison (UK median and sector value columns). Chain volume measures (CVM) are used for comparison of growth of a sector (short-term change and long-term change columns). Time period: 1997-2025 Dataset: Output per job, UK	ONS / 19 May 2026
MFP	Multi-factor productivity, index, 2022=100 Multi-factor productivity (MFP) reflects the overall efficiency with which labour and capital inputs are combined in the production process. The data is released with industries O, P and Q combined, and industries R, S, T and U combined. Therefore, each individual industry uses the value of its combined group. For example, the value for industry R is the value for the combined RSTU group. Time period 1970-2024 Dataset: Multi-factor productivity, annual, UK	ONS / 23 May 2025

Direct Inputs: Employment, Skills & Wellbeing

This category of direct inputs includes a range of indicators that describe the UK's labour market dynamics. These indicators reflect labour quantity, cost and quality.

Indicator	Description	Source / Release
Unit Labour Cost	Unit Labour Cost (ULC), ratio Nominal unit labour costs (ULCs) for an industry are the ratio of total employment costs (not including the costs covered by subsidies) to real gross value added (GVA) for that industry. Time period: 1997-2024 Dataset: Labour cost and labour income, UK	ONS / 15 May 2025
Average Labour Compensation	Average labour compensation per hour worked (ALCH), £/h ALCH for an industry is the total employment costs (not including the costs covered by subsidies) divided by the total number of hours worked in that industry. Time period: 1997-2024 Dataset: Labour cost and labour income, UK	ONS / 15 May 2025
Sector Jobs in Total Employment	Share of jobs by industry in total employment in the UK economy, % Jobs data show how total employment in the economy was distributed across industries. The TPI Productivity Lab estimates Sector Jobs as a percentage of the number of jobs in each industry divided by the total number of jobs in the UK economy. This indicator is not included in the driver categories. It provides context for the sector performance. Time period: 1997-2024 Dataset: TPI Productivity Lab estimates using Output per job, UK	ONS / 15 May 2025
Jobs	Number of Jobs by sector Jobs within each sector is measured using productivity jobs values. It indicates the scale of employment within a sector, providing a snapshot of labour demand at a given point in time. Changes are expressed as percentage changes reflecting that the underlying variable is a level. Time period: 1998-2024 Dataset: TPI Productivity Lab estimates using Output per job, UK	ONS / 15 May 2025

Indicator	Description	Source / Release
High Skills	Proportion of population with high skills, % Represents the proportion of the working-age population (aged 16-64) with high competency skills by industry. High Skills proportion estimates median skill proficiency trends across industries and years. The ONS skills data uses anchor values adapted from the O*NET framework, where each skill is benchmarked against real-world tasks and occupational requirements. Anchor levels 4–7 that map to - advanced analytical or technical processing, strategic, abstract or creative reasoning, and leadership, innovation, or domain expertise - are included in high skills proportion estimates. Time period: 2012-2023 Dataset: <i>TPI Productivity Lab estimates using Skills supply estimates: 2012 to 2023</i>	ONS / 9 Aug 2024
Low Skills	Proportion of population with no or low skills, % The low skills indicator represents the proportion of the working-age population (aged 16-64) with low competency skills by industry. Low Skills proportion estimates median skill proficiency trends across industries and years. The ONS skills data uses anchor values adapted from the O*NET framework, where each skill is benchmarked against real-world tasks and occupational requirements. Level 1–3 that map to routine, observable, simple skill applications, tasks involving basic recall, reaction, or physical coordination, and entry-level or support roles Time period: 2012-2023 Dataset: <i>TPI Productivity Lab estimates using Skills supply estimates: 2012 to 2023</i>	ONS / 9 Aug 2024
Sickness Absence Rate	Sickness absence rate, % of hours lost out of total worked Annual sickness absence rates of workers by industry in the UK labour market (as reported). It represents proportion of total working time lost (hours) due to sickness or injury. It shows how much potential productive labour is being lost. Time period: 2019-2025 Dataset: Sickness absence rate	ONS / 1 May 2026

Direct Inputs: Capital & Investment

These measure capital investment and knowledge inputs. For instance, FDI brings not just capital but also new technologies and management practices, boosting productivity, while GFCF per job is a proxy for capital investment intensity, and an essential input in multifactor productivity. The indicators included in this driver category are critical inputs and context variables that affect sector productivity through capital investment and competitiveness.

Indicator	Description	Source / Release
FDI per Job	Foreign Direct Investment (FDI) by industry per job, £ Captures external capital infusion per employee. Estimated as FDI by industry activity divided by industry jobs to give inward FDI position per job in an economic sector. The value is then converted to real value over time using GDP deflator. Foreign Direct Investment is international investment position in the UK held by foreign companies (inward) by industrial activity. Some of the data in the ONS FDI dataset have been suppressed to prevent disclosure. Industry jobs, taken from the ONS 'Output per Job' publication. Time period: 2021-2024 Datasets: Foreign direct investment involving UK companies: 2024, Output per job, UK, GDP deflators at market prices, and money GDP November 2025	TPI Lab estimates / ONS / FDI: 27 Jan 2026 Jobs: 19 May 2026 Deflator: HM Treasury /27 November 2025
GFCF per Job	Gross fixed capital formation (GFCF) per job, £ Measures capital investment intensity. Estimated as GFCF by industry (CVM £ million) divided by industry jobs from ONS 'Productivity Jobs'. Time period: 1997-2024 Datasets: Annual gross fixed capital formation by industry and asset, Output per job, UK	TPI Lab estimates ONS / GFCF: 13 Nov 2025 Jobs: 19 May 2026
R&D per Job	Real R&D per job by industry, £ Captures capital allocated to innovation activities as levels of R&D expenditure are linked to productivity levels. ONS data on BERD provides a breakdown of R&D performed in UK businesses by industry. Real R&D per job is estimated as R&D expenditure relative to industry jobs. The value is then converted to real value over time using GDP deflator. Time period: 2022-2024 Datasets: Business enterprise research and development, UK, Output per job, UK, GDP deflators at market prices, and money GDP November 2025	TPI Lab estimates ONS / R&D: 28 Nov 2025 Jobs: 19 May 2026 Deflator: HM Treasury /27 November 2025
R&D Intensity	R&D intensity by industry, % Measures R&D investment relative to output. The TPI Productivity Lab estimates R&D intensity as R&D expenditure (£ million) in an industry as % of industry GVA (annual GVA, £million, by section-level industry aggregations). Time period: 2022-2024 Datasets: Business enterprise research and development, UK, Output per hour worked, UK	ONS / R&D: 28 Nov 2025 GVA: 19 May 2026

Indicator	Description	Source / Release
Capital Productivity	Capital productivity by industry, index, 2022=100 Measures output generated per unit of capital services and shows efficiency of capital use in a sector. Across sectors, capital productivity provides insights into how effectively different UK sectors use capital, distinguishing between capital-intensive growth and efficiency-driven performance. Can indicate whether sectors underperform due to capital inefficiency vs. labour or structural issues. Short- and long-term annual percentage change in capital productivity can help track capital-use trends. Time period: 1970-2024 Dataset: Multi-factor productivity, annual, UK	ONS / 23 May 2025
Capital Deepening	Capital deepening by industry, index, 2022=100 Shows capital intensity per hour worked. Indicates higher labour productivity achieved through better availability of capital per worker to invest in technology or machinery to be used by workers. Short- and long-term annual percentage change in capital deepening can help track capital intensity trends. Time period: 1970-2024 Dataset: Multi-factor productivity, annual, UK	ONS / 23 May 2025

Indirect Drivers: Innovation Technology, Market & Structural Dynamics

This driver category shows the role of innovation technology, market exposure and structural dynamics as a group of indirect drivers influencing industry productivity. These are TFP-enhancing strengths. Although not direct production inputs, they are key to explaining productivity differences across economic sectors. Innovation activity, entrepreneurship and the level of business dynamism in an industry are considered crucial elements affecting the economic sector performance. Market exposure, trade openness and the significance of business export activity affects learning, scale economies, and competition, and improve productivity indirectly. Separating these varied indicators from core inputs in the scorecard is consistent with the production function perspective, sharpens the productivity growth signals from each source of productivity gain, and is useful for refining and extending the scorecard in the future, for example by including new additional indicators such as digitalisation level, automation or digital technology adoption.

Indicator	Description	Source / Release
Innovation-active Firms	Innovation-active businesses, % Businesses engaging in innovation activity as % of total businesses by industry. The rate at which new businesses are emerging reflects the extent of entrepreneurial activity in an economic sector. Entrepreneurship, firm dynamism, and firm establishment are identified as significant catalysts for economic wellbeing. The UK Innovation Survey (UKIS) 2023 by ONS/DBT dataset on business demography in the UK provides data for businesses engaging in innovation activity in 2020-2022. The ONS/UKIS consists of a nationally representative sample of businesses with 10 or more employees in sections B-N of the SIC 2007. The data by SIC divisions for manufacturing, transportation, ICT and professional activities sectors is aggregated according to SIC 2007 section level, and the percentage value for the aggregated sections is re-calculated based on the reported weighted sample size of each SIC division included in the respective SIC section. We used the weighted sample size since the UKIS results were based on weighted data for representativeness of the population of firms. Time period: 2008-2024 Datasets: Statistical Annex of UK Innovation Survey Report 2025, 2023, 2021, 2019, 2017, 2015	ONS / Department for Business & Trade (DBT) / 4 June 2026
New Business Births	New businesses by economic sector, % The rate of new enterprises being created is an indicator of the level of entrepreneurial activity by industry. The TPI Lab estimates of new business births (count) as % of all active enterprises (count). Time period: 2019-2024 Dataset: Business demography, UK	ONS / 20 Nov 2025

Indicator	Description	Source / Release
Managerial Skills	Proportion of population with managerial skills Represents the proportion of the working-age population (aged 16-64) with managerial skills by industry. Managerial Skills proportion estimates median skill proficiency trends across industries and years. This measure reflects organisational efficiency and ability to implement productivity-enhancing processes. The ONS skills data uses anchor values adapted from the O*NET framework, where each skill is benchmarked against real-world tasks and occupational requirements. Anchor levels 4–7 used for managerial skills map to the following skill groups: - Cross-Functional Skills: Resource Management, Social Skills, Systems Skills - Generalised Work Activities: Interacting with others, Mental Processes - Knowledge: Business and Management, Communications, Education & Training. Time period: 2012-2023 Dataset: Skills supply estimates: 2012 to 2023	ONS / 9 Aug 2024
Exporters	Exporters of Goods and Services by industry, % Percentage of businesses which are exporters. The TPI Lab estimates of industry exporters as % of all active enterprises. This measure is linked to Export Intensity indicating which UK industry is more export orientated and have access to international markets and competition. Time period: 2011-2024 Dataset: Exporters and importers by industry breakdown (Annual Business Survey)	ONS ABS / 12 Jun 2026
Export Intensity	Exports intensity, % The export intensity of an industry plays a crucial role in driving productivity as companies engaged in global competition typically enhance their efficiency and reduce costs, thereby boosting their productivity levels. Consequently, a higher export performance among firms leads to increased productivity within the industry. This metric is determined by summing the nominal values of trade in goods and services, then dividing the result by the nominal industry GVA value. Because this measurement relies on distinct estimations of goods and services exports, issues related to confidentiality may emerge at the industry level, resulting in gaps within the dataset. Some data cells have been suppressed by the ONS to protect confidentiality. The TPI Data Lab estimates of annual export of goods and services (£ millions) as % of industry GVA (current price, £ millions). Time period: 2016-2023 (for services), 2008-2023 (for goods) Datasets: UK trade in goods by industry, country and commodity, exports; UK trade in services by industry, country and commodity, exports; Output per hour worked, UK	ONS / Trade: 1 May 2025 GVA: 19 May 2026

Indicator	Description	Source / Release
Import Intensity	Import intensity, % Import intensity can indicate how dependent a sector is on foreign inputs for production, and therefore, how exposed it is to global supply chains, price shocks, and international competition. Import intensity and export intensity together provide a balanced picture of a sector's integration in global markets. The TPI Data Lab estimates of annual import of goods and services (£ millions) as % of industry GVA (current price, £ millions). Time period: 2016-2023 (for services), 2008-2023 (for goods) Datasets: UK trade in goods by industry, country and commodity, imports; UK trade in services by industry, country and service type, imports; Output per hour worked, UK	ONS / Trade: 1 May 2025 GVA: 19 May 2026

TPI UK Sectoral Productivity Dashboard

The Sectoral Dashboard allows users to explore the scorecard data in more detail through a range of interactive visualisations. It also incorporates additional data sources that are not used in the scorecards, which are listed below.

Indicator	Description	Source / Release
Labour Productivity (Output per Job) for detailed industries	Output per Job (OpJ) by industry Output per job for each industry is calculated by dividing gross value added (GVA) by the number of jobs in that industry. GVA per job, measured in current prices (CP), is ranked by SIC 2007 division code within each sector for the dashboard. For manufacturing, too many division codes are available to display individually, so the median sub-division within each two-digit division is used instead. Time period: 2009-2023 Dataset: Output per Job for detailed industries	Martin, Josh; Taylor, Clíodhna (2025). Labour productivity estimates for detailed industries in the UK, 2009 to 2023 / 8 October 2025
GVA	Gross Value Added (GVA) by industry for ITL2 areas, current price estimates, £m GVA measures the value a sector adds to the economy, expressed in constant prices (£m) to allow comparison over time. It reflects the scale of economic output after accounting for intermediate inputs and is a core indicator of a sector's economic contribution. Time period: 1998-2023 Dataset: Regional gross value added (balanced) by industry: all ITL regions	ONS / 17 April 2025

Indicator	Description	Source / Release
Labour productivity (output per hour)	Output per Hour (OpH) worked by industry, £/h Output per hour calculated per industry calculated as Gross Value Added (GVA) divided by hours worked in each industry (SIC 2007/NACE section level) Time period: 1997-2024 Datasets: ONS: Output per hour worked, UK Eurostat: Gross value added and income by detailed industry (NACE Rev.2) Employment by detailed industry (NACE Rev.2) - national accounts	ONS / 19 May 2026 Eurostat
Share of hours worked by sector	Hours worked (millions of hours) Hours worked in a sector in the most recent year, as a share of total hours worked in the economy that year Time period: 1997-2024 Datasets: ONS: Output per hour worked, UK Eurostat: Employment by detailed industry (NACE Rev.2) - national accounts	ONS / 19 May 2026 Eurostat